

## COLDPOD · CALCULATION

# ColdPod sizing calculations

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## DOCUMENT CONTROL

|             |                                          |         |                |
|-------------|------------------------------------------|---------|----------------|
| Document ID | CPD-CAL-001                              | Version | 0.1            |
| Status      | Draft                                    | Date    | 2026-09-25     |
| Author      | Amish Chadha                             | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept                   |         |                |
| Repository  | github.com/BoujeeEnjinia1701/coldpod     |         |                |
| Source      | docs/04-calcs/01-sizing.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                       |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 2026-09-25 | Amish Chadha | First issue for TRL 3 (payload, heat leak, PCM, hold times, Peltier and heat sink, refreeze, thermosiphon, freeze fault, logger, mass, cost) |

## ColdPod sizing calculations

On paper, ColdPod meets seven of its seventeen requirements, has five at risk and misses three; two cannot be verified at TRL 3. The insulation is better than the TRL 2 estimate (0.091 W/K rather than 0.11 W/K), so the hold times survive a smaller PCM charge: about 13.0 h with no power at 43 °C, 26.5 h off-grid at 32 °C and 16.5 h off-grid at 43 °C. Those margins are thin against a  $\pm 30$  % uncertainty in the heat leak, so R4, R5 and R6 are at risk. The misses are the freeze fault (R2: a stuck-on driver can pull the liner to about  $-4$  °C past the  $-5$  °C cold-block cut-out), the lid PCM pack (R8: it has no cold path, so the Peltier cannot refreeze it) and mass (R12: 5.60 kg against the relaxed 5.5 kg). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [B9], is the line of that script's output that carries it.

**Safety:** These calculations concern a device meant to protect vaccines and insulin, a 76.8 Wh lithium battery, a combustible paraffin PCM and a heat sink near 53 °C. They are first-principles estimates for a paper proof of concept, not a substitute for datasheets, a fault analysis by a qualified engineer or test. ColdPod is a research and educational prototype, not a medical device, and is not WHO-prequalified. See CPD-PRC-001, Safety.

### ■ Scope and method

The note checks every requirement in CPD-REQ-001 v0.3 against the design in CPD-PRC-001 v0.3 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS` and `levels()`, so the layer dimensions used here are the ones in the STEP files and in drawing CPD-DWG-001. It reads prices from `bom/bom.csv`. Run it from the repo root with `python docs/04-calcs/sizing.py` (about 30 s; the refreeze is a time-stepped simulation).

Status rule for the hold times: **met** if the target holds even with 30 % more heat leak than the nominal estimate; **at risk** if it holds only at the nominal leak; **not met** otherwise.

### ■ Assumptions

*Table 1. Main assumptions.*

| Area                       | Assumption                                                                                                                                                                                                                                            | Basis                                                  |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Ambient cases              | 25 °C clinic, 32 °C hot day, 43 °C WHO hot zone; constant, shaded, no solar gain                                                                                                                                                                      | CPD-REQ-001                                            |
| VIP                        | Aged centre-of-panel conductivity 0.007 W/(m·K), fumed silica core; 25 mm                                                                                                                                                                             | Inside the 0.006 to 0.008 W/(m·K) range usually quoted |
| VIP edges                  | Butt joint between two metallized-film edges with a 1 mm gap: $\psi = 0.015$ W/(m·K) per metre of joint; one edge against foam: 0.0075 W/(m·K)                                                                                                        | Engineering judgment; to confirm with the VIP supplier |
| Geometry of the insulation | Langmuir shape factor for a box: walls A/t, edges 0.54 per metre, corners 0.15 t                                                                                                                                                                      | Standard conduction shape factors                      |
| Pipe crossing              | 20 mm polyurethane foam strip (0.024 W/(m·K)) across the +X wall; two 8 mm tubes with a stainless section, 0.3 mm wall, 40 mm long                                                                                                                    | Model                                                  |
| Sensor wires               | Six 28 AWG copper conductors, 100 mm path along a VIP joint                                                                                                                                                                                           | Payload and liner probes                               |
| PCM                        | RT 5 HC class; 85 % pouch fill; 770 kg/m <sup>3</sup> liquid, 880 kg/m <sup>3</sup> solid; 180 kJ/kg usable inside 2 to 8 °C (of about 250 kJ/kg listed over the full range); solid conductivity 0.2 W/(m·K); cp 2.0 kJ/(kg·K) liquid, 1.8 solid      | Rubitherm RT range data, derated                       |
| Hold end point             | Hold ends when the jacket or the lid pack is fully melted, whichever comes first; the sensible heat of the payload and liner is kept as unclaimed margin                                                                                              | Conservative                                           |
| Peltier module             | 127-couple modules, V <sub>max</sub> 15.4 V, $\Delta T_{max}$ 67 K at a 300 K hot side; I <sub>max</sub> 3, 4 or 6 A (TEC1-12703, 12704, 12706 classes); Seebeck coefficient, resistance and conductance derived from those ratings and held constant | Standard derivation from datasheet maxima              |
| Cold path                  | 0.40 K/W from the can to the module cold face (loop thermosiphon, cold block, interface)                                                                                                                                                              | Engineering judgment for 8 mm two-phase tubes          |
| Hot path                   | 0.55 K/W from the module hot face to air (80 x 80 x 30 mm finned sink with a 70 mm fan, 0.50 K/W, plus 0.05 K/W interface)                                                                                                                            | Typical for this sink class; to confirm                |
| Control                    | Powered hold keeps the evaporator can at 2 °C; refreeze floor –2 °C at the can; module fed smooth DC from a buck driver, 90 % efficient; fan 0.6 W, controller 0.1 W                                                                                  | Design intent                                          |
| Battery                    | 76.8 Wh; 85 % (65.3 Wh) available to cooling, 15 % kept for the logger                                                                                                                                                                                | CPD-PRC-001                                            |
| Refreeze                   | 25 W cap to the module and 12 W to the battery while charging, inside a 45 W USB-C PD budget; can to liquid PCM 40 W/(m <sup>2</sup> ·K) through the pouch film                                                                                       | Design intent                                          |
| Lid pack                   | Couples to the cavity through about 12 mm of stably stratified air, 2 W/(m <sup>2</sup> ·K)                                                                                                                                                           | Conduction through still air                           |
| Mass                       | PETG 1.27 g/cm <sup>3</sup> ; printed walls with 2.7 mm of solid perimeters and 20 % infill for the rest; VIP 190 kg/m <sup>3</sup> including film; LiFePO <sub>4</sub> 32700 cell 0.14 kg                                                            | Model volumes; typical values                          |

## ■ A. Payload and size (R3, R13)

The liner is 172 x 107 x 78.5 mm inside [A1]. Less the printed rack and the buffered probe, the usable volume is 1.36 L [A2]. It holds 24 pens of 16 mm diameter, in four layers of six [A3], using 100 of 107 mm across and 66 of 78.5 mm in height [A4]; pens up to 172 mm long fit [A5]. R3 is met.

Overall, with the handle up, the case is 368 x 214 x 207 mm (14.5 x 8.4 x 8.1 in) [A6, A7], inside the 400 x 250 x 250 mm envelope of R13: met.

## ■ B. Heat leak (drives R4 to R7)

The inside of the insulation is 205 x 140 x 110 mm [B1]. With Langmuir shape factors of 4.809 m for the body and 1.536 m for the lid [B2], the centre-of-panel conductance is 0.0337 W/K and 0.0107 W/K [B3]. The other paths are:

Table 2. Heat-leak paths.

| Path                                           | Conductance (W/K)               | Tag |
|------------------------------------------------|---------------------------------|-----|
| Body panels, centre of panel                   | 0.0337                          | B3  |
| Lid plug, centre of panel                      | 0.0107                          | B3  |
| Body VIP joints, 1.53 m                        | 0.0229                          | B4  |
| Lid plug junction                              | 0.0103                          | B5  |
| Foam strip at the pipe crossing                | 0.0056                          | B6  |
| Thermosiphon stainless sections (reverse path) | 0.0058                          | B7  |
| Sensor wires                                   | 0.0019                          | B8  |
| <b>Body / lid / total</b>                      | <b>0.0700 / 0.0211 / 0.0911</b> | B9  |

With the PCM at 5 °C, the heat leak is 1.82 W at 25 °C, 2.46 W at 32 °C and 3.46 W at 43 °C [B10]. The TRL 2 estimate was 0.11 W/K [B11]; the explicit edge and joint terms come to less than the 1.8 multiplier it used. Joints and edges are still about 45 % of the total, so the supplier's edge data matter.

## ■ C. Phase-change material

The aluminium evaporator can (section G) and the loop tubes take some of the PCM space. The jacket holds 1.055 L and the lid pack 0.430 L [C1], which at 85 % fill is 0.691 kg and 0.282 kg, 0.972 kg in all [C2]. At 180 kJ/kg that is 124 kJ in the jacket and 51 kJ in the lid, 175 kJ in all [C3], or 48.6 Wh [C4]. TRL 2 assumed 1.06 kg and 191 kJ.

## ■ D. Passive hold (R4)

Table 3. Hold with no power, starting with all the PCM frozen.

| Ambient | Jacket melts (h) | Lid pack melts (h) | Pooled (h) | Tag |
|---------|------------------|--------------------|------------|-----|
| 25 °C   | 24.7             | 33.4               | 26.7       | D1  |
| 32 °C   | 18.3             | 24.7               | 19.8       | D1  |
| 43 °C   | <b>13.0</b>      | 17.6               | 14.0       | D1  |

The jacket melts first, so R4 is 13.0 h at 43 °C [D2] against 12 h. With 30 % more heat leak it falls to 10.0 h [D3]; an 8 % rise in heat leak uses up the margin [D4]. R4 is **at risk**.

## ■ E. Peltier module, cold path and heat sink (R7)

A Peltier module that is switched off conducts heat. The derived off-state conductance is 0.268 W/K for the TEC1-12703 class, 0.357 W/K for the 12704 and 0.536 W/K for the 12706 [E1]. Even the smallest is 2.9 times the whole box's heat leak [E2], which is why the thermosiphon diode (section H) is needed.

The same conductance decides which module can hold at 43 °C, because the hot side runs about 50 K above the cold side:

*Table 4. Powered hold, can at 2 °C: current, module power, COP and input power (module through a 90 % driver, plus fan and controller).*

| Module     | 25 °C                             | 32 °C                             | 43 °C                               | Tag |
|------------|-----------------------------------|-----------------------------------|-------------------------------------|-----|
| TEC1-12703 | 0.73 A, 3.1 W, 0.67, <b>4.2 W</b> | 1.03 A, 6.1 W, 0.45, <b>7.5 W</b> | 1.69 A, 15.9 W, 0.23, <b>18.4 W</b> | E3  |
| TEC1-12704 | 0.93 A, 3.9 W, 0.54, 5.0 W        | 1.31 A, 7.6 W, 0.36, 9.2 W        | 2.21 A, 21.0 W, 0.18, 24.1 W        | E3  |
| TEC1-12706 | 1.34 A, 5.5 W, 0.38, 6.8 W        | 1.94 A, 11.4 W, 0.24, 13.3 W      | Cannot lift the load                | E3  |

The TEC1-12706 named at TRL 2 cannot hold at 43 °C with this heat sink: its own back-conduction exceeds what it can pump. The **TEC1-12703 class** is chosen. At 43 °C its cold face is at 0.5 °C and its hot face at 53.8 °C [E4]; it needs 9.5 V [E5], and the sink rejects 5.2 W, 8.9 W and 19.7 W at 25, 32 and 43 °C [E6]. The sink base reaches about 53 °C at 43 °C [E7].

R7 asks for an indefinite hold up to 43 °C on 12 V or 45 W USB-C PD. The input is 18.4 W, well inside 41.4 W available from a 45 W PD source [E8], and 1.94 A from a 10 V vehicle input [E9]. The module needs 9.5 V, so a low vehicle supply of 10 V leaves little headroom for a buck driver; a buck-boost stage may be needed. The cooling side of R7 is met, but the lid pack melts after about 21 h of powered hold at 43 °C (section F), after which the top layer of the payload is at risk. R7 is **at risk**.

The driver matters. Unfiltered on/off pulse-width modulation at 12.8 V, giving the same average heat lift at 43 °C, needs a 0.79 duty and 25.6 W at the module [E10], 61 % more than smooth DC (15.9 W). The power board therefore uses a buck driver with an LC filter, not on/off PWM. CPD-PRC-001 v0.2 said PWM improves COP; that is corrected in v0.3.

## ■ F. Off-grid hold (R5, R6)

65.3 Wh of the battery is available for cooling [F1]. While the battery runs, the module holds the can at 2 °C and the PCM stays frozen; the lid pack melts slowly, because its heat path bypasses the can. Then the PCM holds passively.

*Table 5. Battery, then PCM.*

| Ambient | Battery phase (h) | Passive phase (h) | Total (h)   | With 30 % more leak (h) | Tag     |
|---------|-------------------|-------------------|-------------|-------------------------|---------|
| 25 °C   | 15.6              | 22.1              | 37.6        |                         | F2      |
| 32 °C   | 8.7               | 17.8              | <b>26.5</b> | 20.1                    | F2, F2b |
| 43 °C   | 3.5               | 13.0              | <b>16.5</b> | 12.4                    | F2, F2b |

R5 (24 h at 32 °C) and R6 (16 h at 43 °C, relaxed by CPD-DDR-001 D5) are met at the nominal heat leak and missed at +30 %: both are **at risk**. For reference, the 153.6 Wh 4S2P pack not chosen in D4 would give 18.6 h at 43 °C [F6].

In powered hold at 43 °C the lid pack absorbs a net 0.69 W [F3] and is fully melted after about 21 h [F4]. Once melted it settles near 13.8 °C [F5], with only a 12 mm air gap between it and the top layer of pens, which then sit somewhere between the liner (about 3 °C) and the pack. This is why R1 and R7 are at risk in long hot holds.

## ■ G. Refreeze (R8)

The decided evaporator position (outer face of the PCM, CPD-DDR-001 D3) is sized here. A 90 x 60 mm plate on one end of the jacket, as drawn at TRL 2, cannot freeze a paraffin jacket that runs 200 mm around the liner: solid paraffin conducts only about 0.2 W/(m·K). The evaporator is therefore a 1.0 mm aluminium can that lines the outer face of the jacket on four sides and the floor, 0.0943 m<sup>2</sup> in contact [G1], with an 8 mm copper evaporator loop around its base. The frozen jacket is effectively 8.3 mm thick [G2], and the can wall above the loop acts as a fin with 0.69 efficiency [G3].

A time-stepped model (sensible cooling of the liquid PCM through the pouch film, then a quasi-steady freezing front, with the module current chosen each step within the 25 W cap and the –2 °C can floor) gives:

Table 6. Jacket refreeze from 25 °C, box empty, 25 °C ambient.

| Module     | Sensible part (h) | Total (h) | Tag |
|------------|-------------------|-----------|-----|
| TEC1-12703 | 0.9               | 5.5       | G4  |
| TEC1-12704 | 0.9               | 5.3       | G4  |
| TEC1-12706 | 0.8               | 5.5       | G4  |

Conduction through the freezing PCM alone would allow about 1.6 h [G5], so the module, not the PCM, sets the rate. The jacket meets the 8 h target.

The lid pack has no cold path to the evaporator: only the air gap and the lid gasket [G6]. The Peltier cannot refreeze it, so **R8 is not met** for the whole PCM, and the TRL 2 hold times quietly assumed a lid pack frozen some other way. Two fixes are sized for review (not adopted): an aluminium plate under the lid pack that seats on the can rim when the lid closes, which would refreeze the lid pack in about 2.3 h [G7] and add 0.12 kg [O3]; or no lid pack and a 20 mm jacket holding 1.010 kg, all actively frozen [N1], with the body 10 mm longer and wider and 10 mm lower [N2].

The battery recharges in about 6.7 h at 12 W alongside the refreeze [G8], for a total input of 40.5 W [G9], inside the 45 W PD budget.

## ■ H. Thermosiphon: diode ratio and tilt

The loop thermosiphon conducts about 2.50 W/K forward and 0.0058 W/K in reverse (the stainless wall sections), a ratio of 431 [H1]. For comparison, bolting the chosen module to the liner would raise the box conductance to 0.359 W/K [H2] and cut the passive hold at 43 °C to 3.6 h [H3]. The thermosiphon keeps its place (CPD-DDR-001 D2).

The condenser sits 75.0 mm above the evaporator loop [H4]. The whole loop stays below the condenser for tilts up to 18° with the cooling head down [H5] and 41° with the front or back down [H7]; with the cooling head down, part of the loop stays below the condenser up to 60° [H6]. Beyond those angles the part of the loop above the condenser dries out and cooling falls; the PCM carries the load meanwhile. Whether a motorbike carrier holds within 18° on average is an open question for a lab or field test after TRL 3.

## ■ I. Freeze fault (R2)

In normal operation the coldest surface (the can, down to  $-2^{\circ}\text{C}$  during refreeze) is separated from the liner by the PCM, and the controller stops cooling at a liner reading of  $3^{\circ}\text{C}$ , so R2 holds. With a stuck-on driver, the module draws about 3.6 A from the 14.4 V charge rail [I4] until the hardware cut-out on the cold block opens at  $-5^{\circ}\text{C}$  [I1]. Once the PCM is fully frozen, nothing stops the liner from following the can down to about  $-4^{\circ}\text{C}$  in a cool room [I2]. The liner and payload alarms would sound, but R2 asks that no payload-contact surface fall below  $1^{\circ}\text{C}$  even in this fault. **R2 is not met.**

A second hardware cut-out on the liner, in series with the first, fixes it. A lumped estimate of the liner after the cut-out opens (the payload and liner sharing heat with the colder frozen jacket) gives  $1.2^{\circ}\text{C}$  for a  $2^{\circ}\text{C}$  setting and  $2.0^{\circ}\text{C}$  for a  $3^{\circ}\text{C}$  setting [I3]. A  $3^{\circ}\text{C}$  liner cut-out is proposed, awaiting Amish.

## ■ J. Logger (R9, R11)

Sixty days of one 16-byte record per minute take 1.38 MB of the 2.10 MB flash [J1]: R9 is met on storage, and on accuracy by sensor selection. The logger and standby loads come to about 0.78 mW [J2]; with a 5 mW design allowance [J3], the 15 % reserve lasts about 96 days [J4], far beyond the 14 days of R11: met.

## ■ K. Mass (R12)

Table 7. Empty mass [K1].

| Part                       | kg    | Part                                 | kg    |
|----------------------------|-------|--------------------------------------|-------|
| Shell (PETG, printed)      | 0.740 | Loop tubes (copper)                  | 0.089 |
| Lid cap, gasket, latches   | 0.357 | Cold block (aluminium)               | 0.071 |
| Handle and strap           | 0.200 | Peltier module                       | 0.025 |
| VIP set                    | 0.860 | Heat sink                            | 0.197 |
| PCM                        | 0.972 | Fan and guard                        | 0.060 |
| PCM pouches                | 0.080 | End housings (PETG, printed)         | 0.404 |
| Liner (aluminium)          | 0.257 | LiFePO <sub>4</sub> cells, 4 x 32700 | 0.560 |
| Rack (PETG)                | 0.083 | BMS, fuse, holder                    | 0.080 |
| Evaporator can (aluminium) | 0.254 | Electronics                          | 0.160 |
|                            |       | Wiring and hardware                  | 0.150 |

The total is **5.60 kg (12.3 lb)** empty [K2, K3] and 6.28 kg with 24 pens [K4], against the relaxed target of 5.5 kg: **R12 is not met** by 0.10 kg. The evaporator can (0.25 kg), which TRL 2 did not have, is most of the growth. Thinner printed parts (3 mm shell, 5 mm lid cap, 2 mm housings) would save 0.06, 0.04 and 0.11 kg [O1], for 5.39 kg [O2]; with the lid cold plate (0.12 kg, [O3]) the case would be about 5.51 kg.

## ■ L. Cost and battery (R14, R15)

All 16 BOM lines are priced [L1]; the total is \$295 [L2] against the \$300 budget set by CPD-DDR-001 D10: met, with a \$5 margin and no custom PCB. The battery is 76.8 Wh [L3], under the 100 Wh airline limit: R14 is met.

## ■ P. Foam variant (CPD-DDR-001 D7)

The documented low-cost variant replaces the VIPs with 25 mm polyurethane foam ( $0.024 \text{ W}/(\text{m}\cdot\text{K})$ ). With the same shape factors and no VIP joints, the conductance is  $0.160 \text{ W/K}$  [P1] and the passive hold at  $43^\circ\text{C}$  falls to about 7.4 h [P2], so the foam variant does not meet R4. TRL 2 estimated about 7 h.

## ■ Checks against earlier figures

*Table 8. TRL 2 figures in CPD-PRC-001 v0.2 and the README, checked against this note. The docs are corrected in v0.3.*

| Quantity                            | TRL 2                               | TRL 3                                                                                          | Tag       |
|-------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------|-----------|
| Overall conductance                 | 0.11 W/K                            | 0.091 W/K                                                                                      | B9        |
| PCM                                 | 1.06 kg, 191 kJ                     | 0.972 kg, 175 kJ                                                                               | C2,<br>C3 |
| Payload volume                      | about 1.2 L                         | 1.36 L                                                                                         | A2        |
| Passive hold 25 / 32 / 43 °C        | 24 / 18 / 12.7 h                    | 24.7 / 18.3 / 13.0 h                                                                           | D1        |
| Input power to hold 25 / 32 / 43 °C | 4.7 / 8.1 / 17.4 W                  | 4.2 / 7.5 / 18.4 W                                                                             | E3        |
| Battery then PCM 25 / 32 / 43 °C    | 38 / 26 / 16 h                      | 37.6 / 26.5 / 16.5 h                                                                           | F2        |
| Refreeze                            | about 5 h (text), about 6 h (table) | 5.5 h, jacket only                                                                             | G4        |
| Heat rejected 25 / 32 / 43 °C       | 6 / 10 / 21 W                       | 5.2 / 8.9 / 19.7 W                                                                             | E6        |
| Heat sink temperature at 43 °C      | 55 to 65 °C                         | about 53 °C                                                                                    | E7        |
| Peltier off-state conductance       | 0.3 to 0.5 W/K                      | 0.27 to 0.54 W/K                                                                               | E1        |
| Peltier module                      | TEC1-12706                          | TEC1-12703 class                                                                               | E3        |
| PWM "for better COP"                | stated                              | smooth DC; on/off PWM costs 61 % more                                                          | E10       |
| Battery recharge                    | about 3 h at 25 W                   | about 6.7 h at 12 W alongside refreeze                                                         | G8        |
| Logger reserve                      | weeks                               | about 96 days                                                                                  | J4        |
| Log storage, 60 days                | about 1.4 MB                        | 1.38 MB                                                                                        | J1        |
| Size                                | 370 x 215 x 205 mm                  | 368 x 214 x 207 mm                                                                             | A6        |
| Mass empty                          | 5.2 kg                              | 5.60 kg                                                                                        | K2        |
| Parts cost                          | \$280                               | \$295                                                                                          | L2        |
| Warm-payload pull-down              | about 3.7 h                         | not recomputed; removed from the precis, which keeps the advice to load the payload pre-cooled |           |

## ■ Results

Table 9. Requirements against this note. Status: met, at risk, not met, or not verifiable at TRL 3.

| ID  | Target (CPD-REQ-001 v0.3)                                                | Value                                                                                         | Status                  |
|-----|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------|
| R2  | No payload-contact surface below 1 °C, including a stuck-on driver       | Normal: met. Fault: liner near –4 °C with the –5 °C block cut-out [I2]                        | Not met                 |
| R8  | Refreeze a fully melted PCM in 8 h or less at 25 °C                      | Jacket 5.5 h [G4]; lid pack has no cold path [G6]                                             | Not met                 |
| R12 | 5.5 kg or less empty (relaxed)                                           | 5.60 kg [K2]                                                                                  | Not met                 |
| R1  | 2 to 8 °C at the payload probe                                           | Can at 2 °C, liner 2 to 5 °C; lid path warms the top layer in long powered hold at 43 °C [F5] | At risk                 |
| R4  | 12 h or more at 43 °C, no power                                          | 13.0 h; 10.0 h at +30 % leak [D2, D3]                                                         | At risk                 |
| R5  | 24 h or more at 32 °C, battery then PCM                                  | 26.5 h; 20.1 h at +30 % leak [F2]                                                             | At risk                 |
| R6  | 16 h or more at 43 °C, battery then PCM (relaxed)                        | 16.5 h; 12.4 h at +30 % leak [F2]                                                             | At risk                 |
| R7  | Hold indefinitely up to 43 °C on 12 V or 45 W USB-C PD                   | 18.4 W input [E8]; lid pack melts after about 21 h [F4]                                       | At risk                 |
| R3  | 1.0 L or more; 20 or more pens up to 170 mm                              | 1.36 L; 24 pens; up to 172 mm [A2 to A5]                                                      | Met                     |
| R9  | ±0.5 °C, log every 1 min, 60 days, CSV export                            | 1.38 MB of 2.10 MB [J1]; accuracy by selection                                                | Met                     |
| R11 | Logger runs 14 days after cooling stops                                  | About 96 days [J4]                                                                            | Met                     |
| R13 | Fits 400 x 250 x 250 mm                                                  | 368 x 214 x 207 mm [A6]                                                                       | Met                     |
| R14 | Battery 100 Wh or less                                                   | 76.8 Wh [L3]                                                                                  | Met                     |
| R15 | Parts \$300 or less; no custom PCB                                       | \$295 [L2]; modules on perfboard                                                              | Met                     |
| R17 | "Research prototype, not a medical device" on box, start screen and logs | By design                                                                                     | Met                     |
| R10 | Alarm logic and thresholds                                               | Design intent only; no firmware sketch exists at TRL 3                                        | Not verifiable at TRL 3 |
| R16 | IP54 bays; 0.5 m drop while loaded                                       | Needs a test                                                                                  | Not verifiable at TRL 3 |

Counts: 7 met, 5 at risk, 3 not met, 2 not verifiable at TRL 3.

## ■ Limits of this note

- All values are paper estimates. The heat leak (joint  $\psi$ , aged VIP conductivity), the thermosiphon resistance, the heat sink resistance and the PCM's usable latent heat inside 2 to 8 °C are the largest uncertainties, in that order.
- The Peltier model uses constant properties derived from datasheet maxima. Real modules vary by maker.
- The lid pack and payload temperatures in long powered holds come from a two-node estimate; stratification and contact with the rack are not modelled.
- No solar gain is included. A carrier in direct sun would see more heat than the 43 °C shaded case.